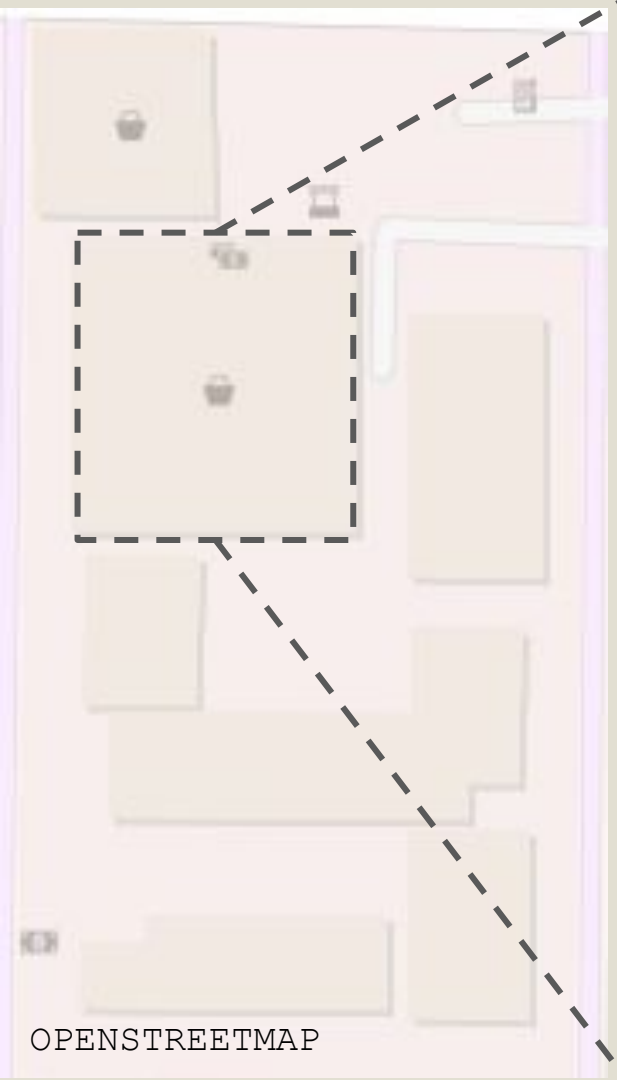
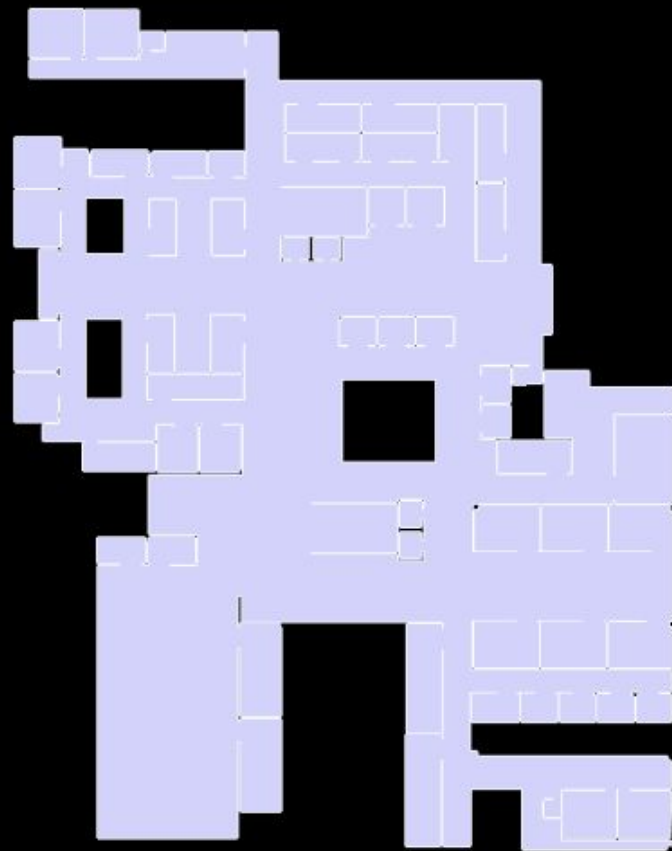


ASSIGNMENT_02

SPATIAL ANALYSIS OF SHOPPING MARKET
JAYANAGAR, BENGALURU
INDIA



OPENSTREETMAP



Ground_floor_plan

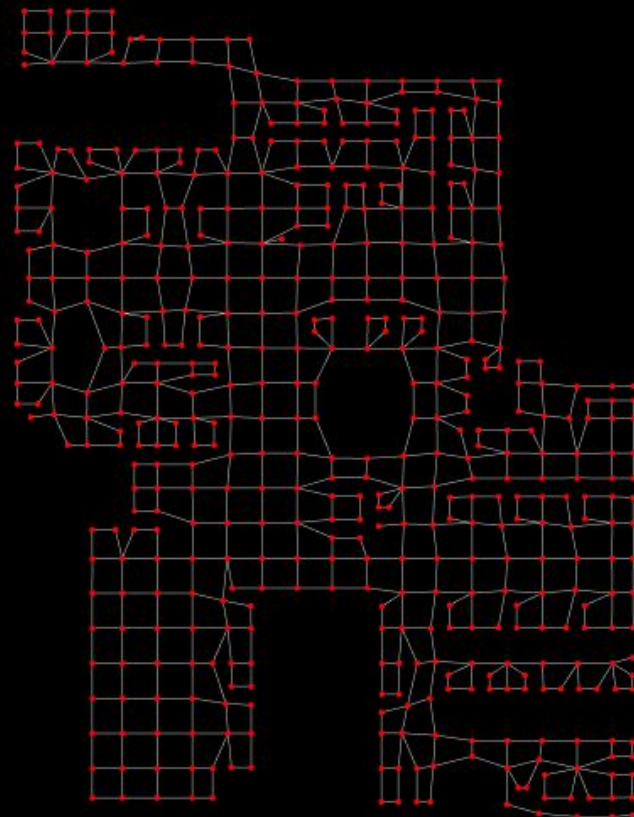
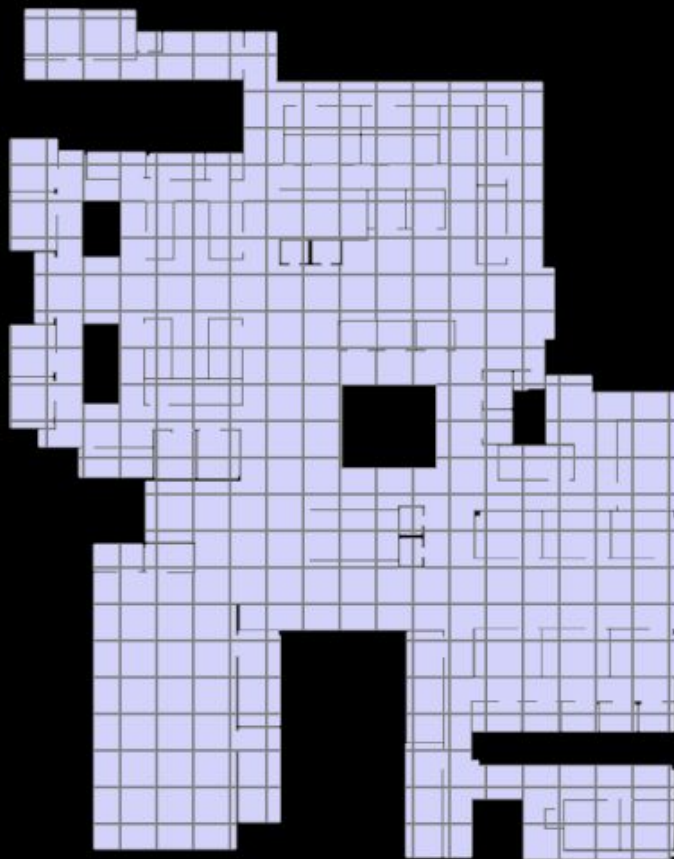
The analyzed building complex covers approximately 14,624 m² and consists of multiple interconnected blocks with typical 4-7 floor structures organized around central circulation zones.

GRID OVERLAY

#Grid_sizes

uRange = 6m

vRange = 6m



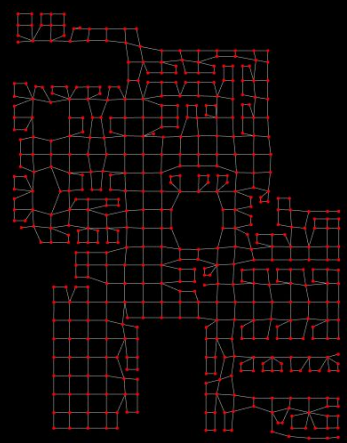
A grid size of 6 m
for both U and V
ranges was
selected because
smaller grid
intervals
significantly
increased the
computational time
during the spatial
and visibility
analysis process.

SHORTEST PATH

#Grid_sizes

uRange = 6m

vRange = 6m



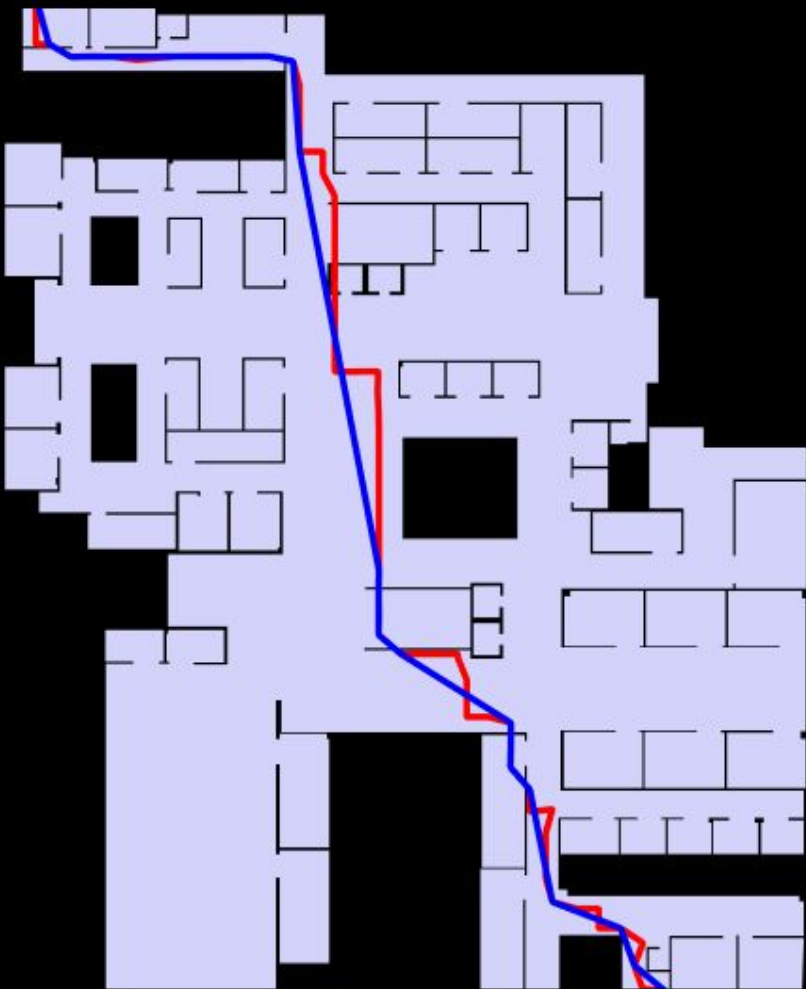
#Shortest Path

Original SP Length: 229.39

Straightened SP Length: 206.21

Number of edges in path: 80

Number of turns: 79



CLOSENESS CENTRALITY / INTEGRATION

#Grid_sizes

uRange = 6m

vRange = 6m

#Closeness Centrality Statistics

Min : 0.0431

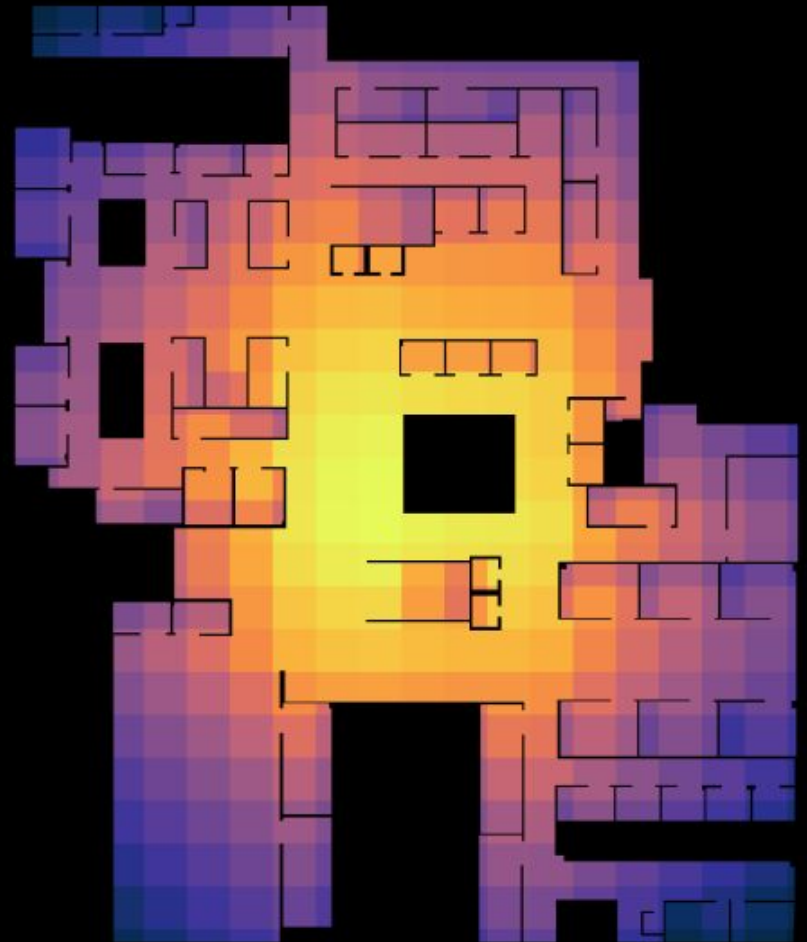
Max : 0.0963

Mean: 0.0694

Spaces analysed: 511

#Key Observation

The closeness centrality analysis shows that the central circulation spaces are the most spatially integrated and accessible areas of the market, while peripheral zones are more segregated and less connected.



BETWEENNESS CENTRALITY

#BETWEENNESS CENTRALITY

Min : 0.0
Max : 0.1817
Mean: 0.0273

TOP 5 MOST TRAVERSED SPACES

1. Vertex at (-4.41, 18.45) --> BC: 0.1817
2. Vertex at (30.21, -0.02) --> BC: 0.1798
3. Vertex at (30.44, 6.55) --> BC: 0.1672
4. Vertex at (30.59, -23.46) --> BC: 0.1666
5. Vertex at (30.93, 12.23) --> BC: 0.1585

LEAST TRAVERSED SPACES

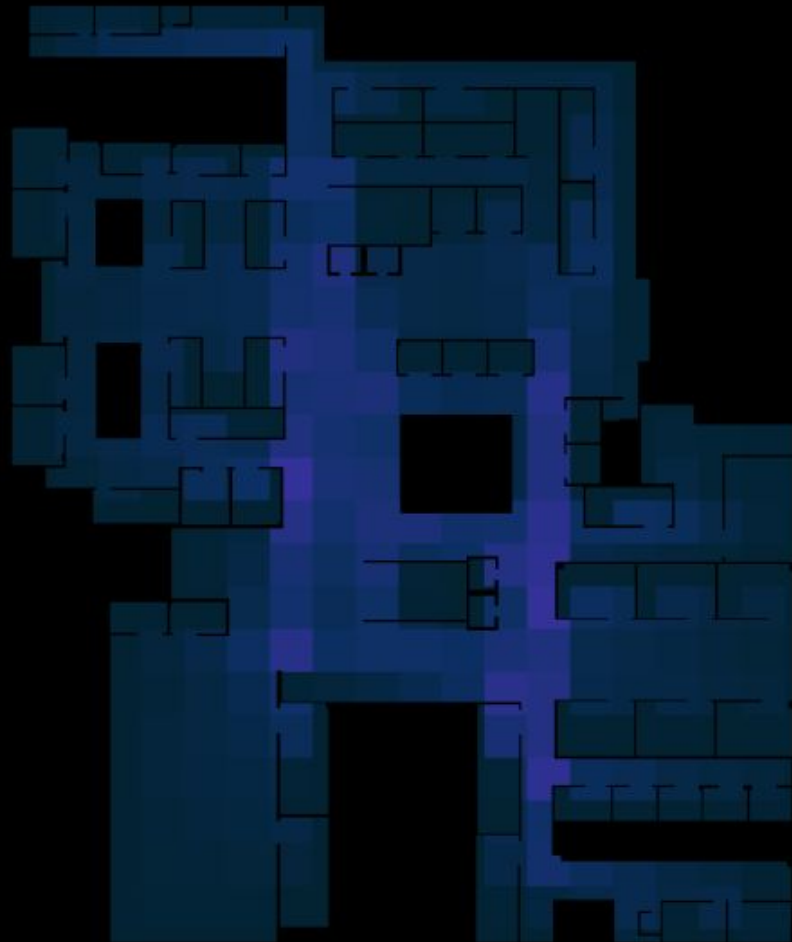
1. Vertex at (64.41, -22.77) --> BC: 0.0
2. Vertex at (-19.7, 83.4) --> BC: 0.0
3. Vertex at (64.63, 23.71) --> BC: 0.0
4. Vertex at (64.4, -49.79) --> BC: 0.0
5. Vertex at (-39.85, 78.76) --> BC: 0.0

HOTSPOTS
Spaces above average BC: 168

CRITICAL CONNECTOR SPACES
High-flow bottlenecks: 92

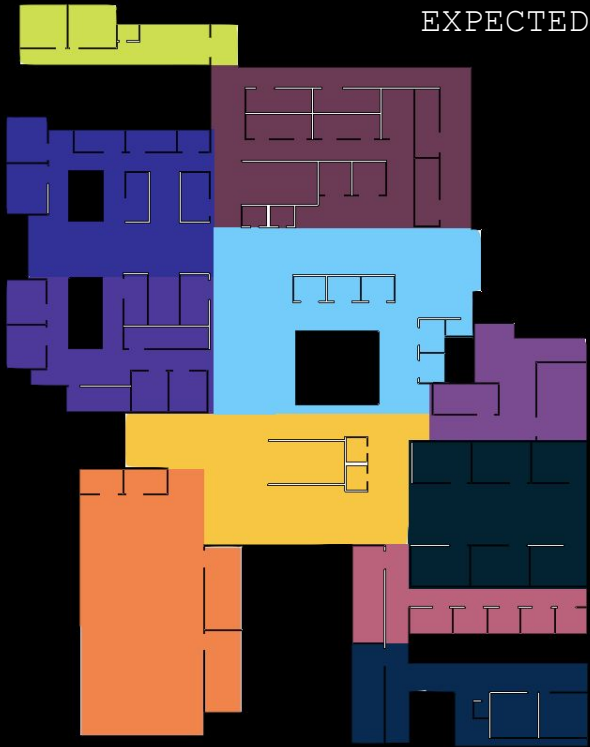
#Key Observation

The betweenness centrality analysis highlights the main circulation corridors that act as critical connectors within the market, carrying the highest movement flow between different spatial zones.

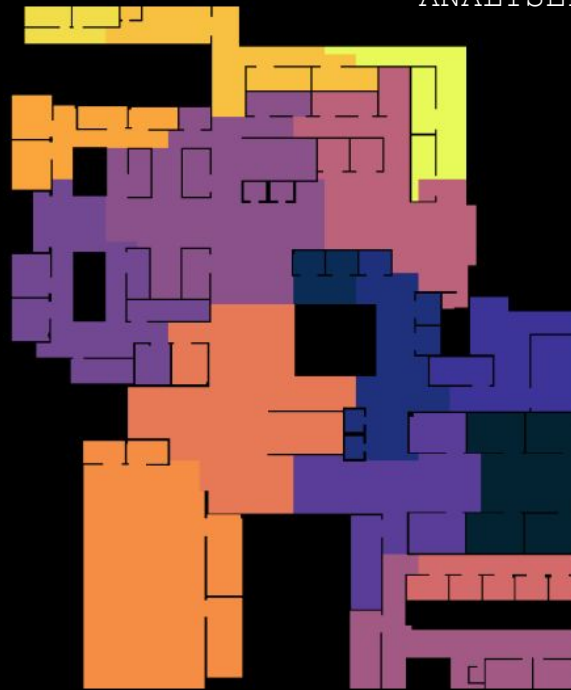


COMMUNITY DETECTION

EXPECTED



ANALYSED



Expected

The **Expected** diagram represents the manually interpreted zoning of the market:

- retail clusters
- circulation-based groupings
- larger anchor spaces
- separated wings of the building

These are the spatial communities anticipated from visual observation of the plan.

Analysed

The **Analysed** diagram shows the communities generated automatically using:

- graph connectivity
- adjacency relationships
- shortest path behaviour
- spatial network structure

Each color represents one detected community within the market network.

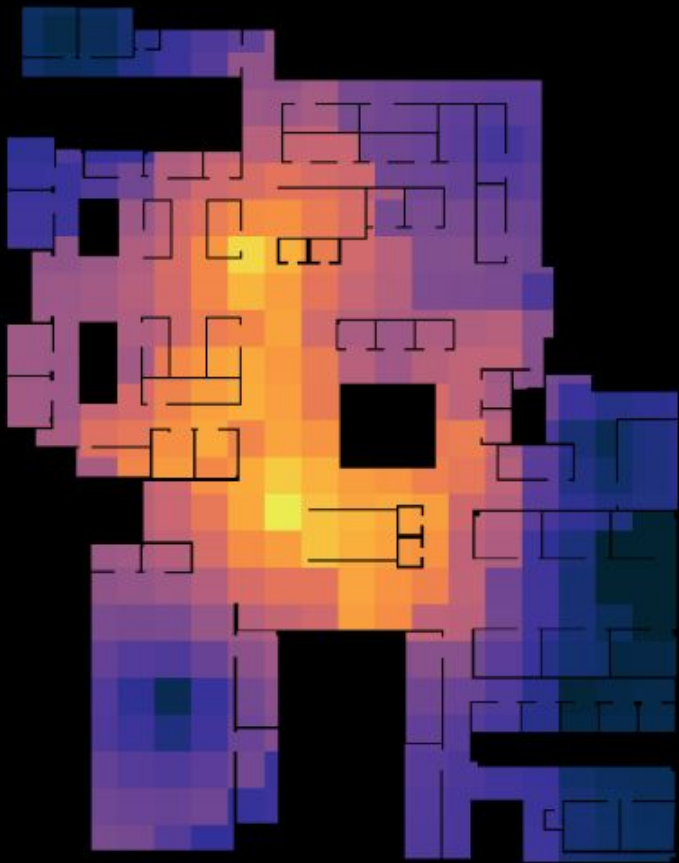
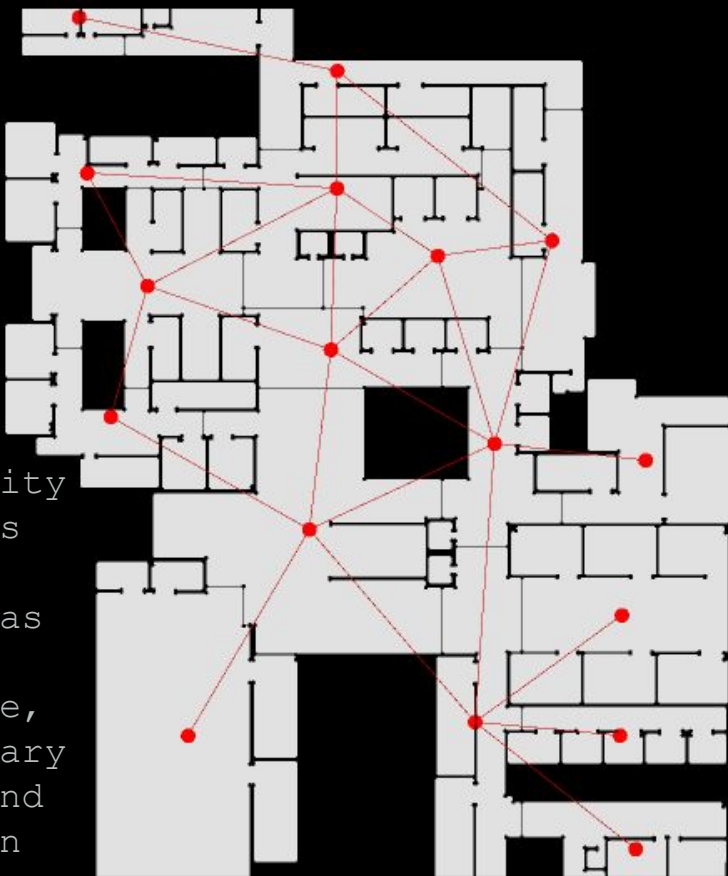
DEGREE OF CENTRALITY

DEGREE CENTRALITY

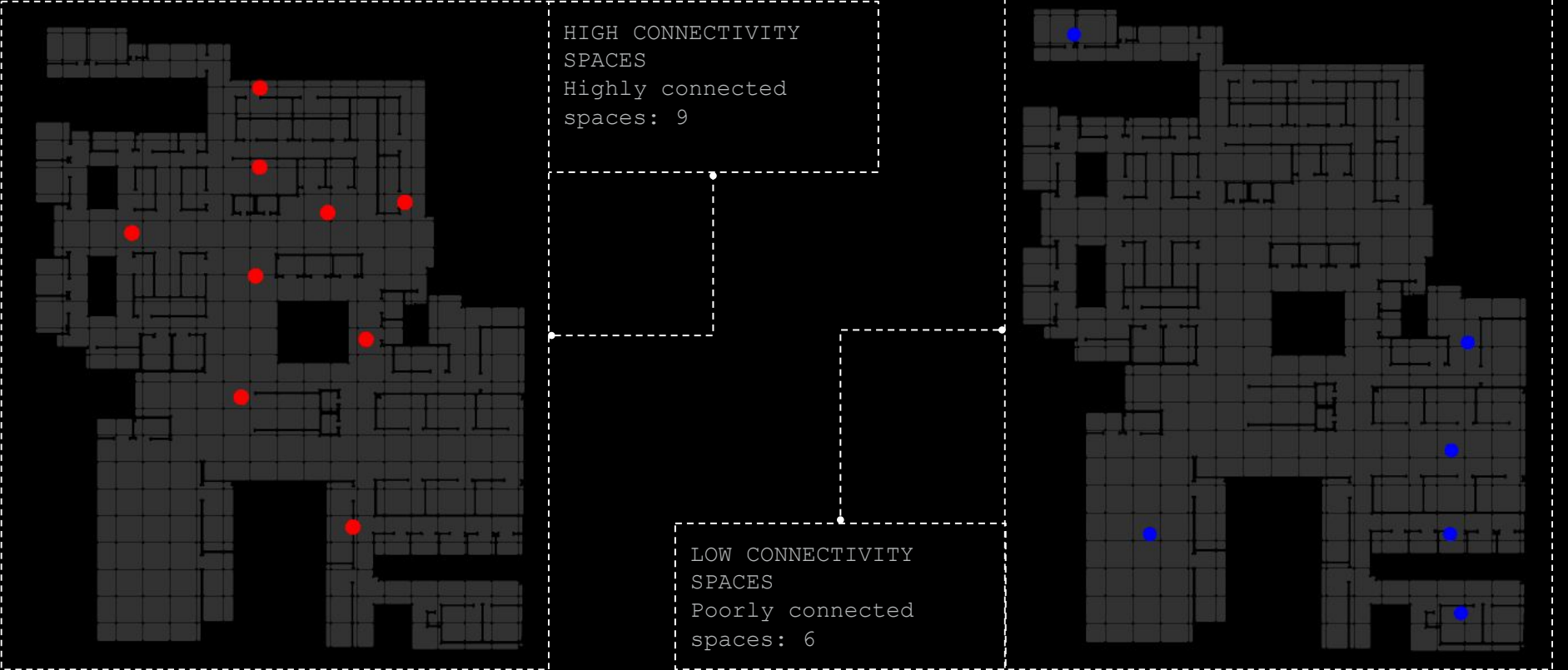
Min : 0.0625
Max : 0.375
Mean: 0.1838
Spaces analysed:
17

#Key Observation

The degree centrality analysis identifies the central circulation spine as the most connected and accessible zone, acting as the primary hub for movement and spatial integration within the layout.



DEGREE OF CENTRALITY



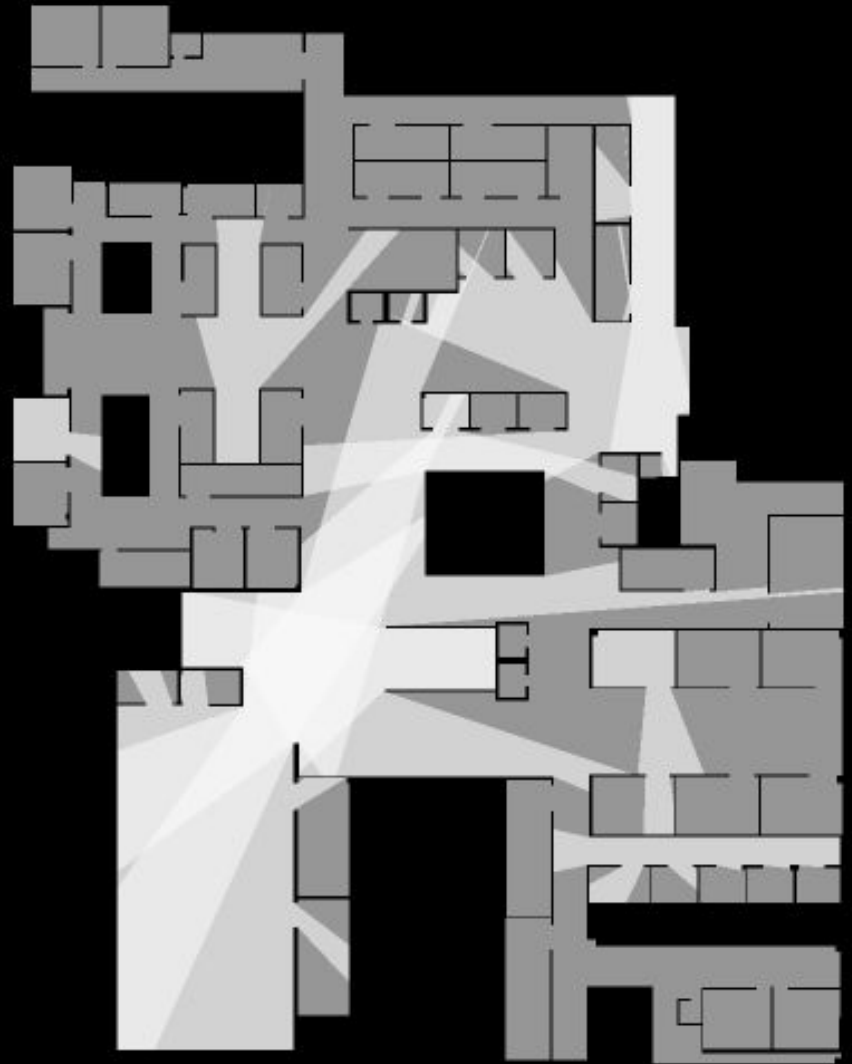
#Key Observation

The analysis reveals a highly connected central circulation spine that functions as the primary movement hub, while peripheral edge spaces remain comparatively isolated with low spatial connectivity.

VISIBILITY GRAPH ANALYSIS

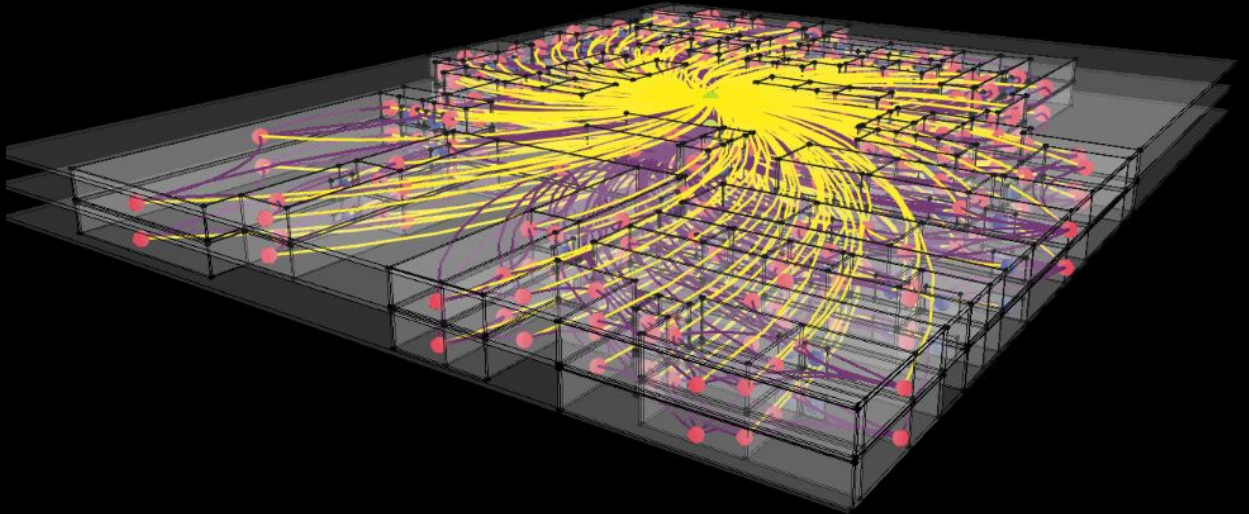
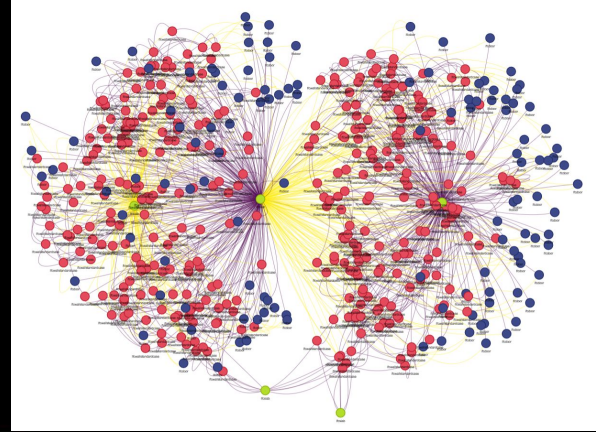
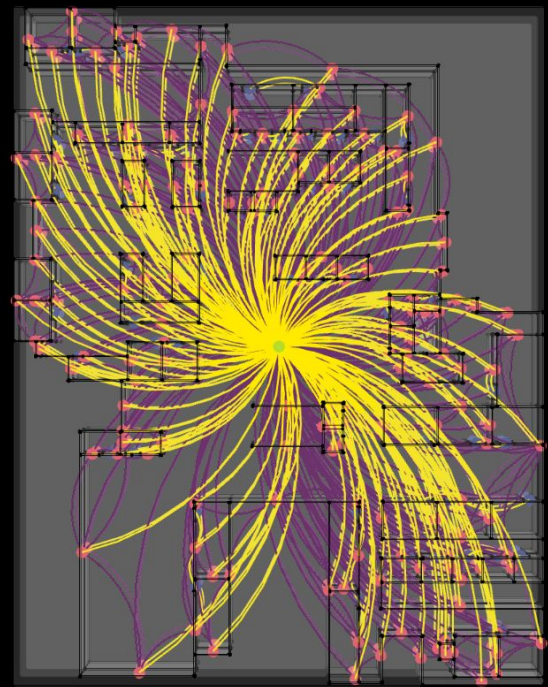
#Key Observation

The visibility graph shows that the central circulation spine has the highest visual connectivity, while peripheral rooms and corner spaces remain visually isolated with limited lines of sight.



Spatial Relationships

(IFC FILE)



#Key Observation

The spatial relationship analysis reveals a highly centralized connectivity structure, where a few dominant hub spaces control and distribute movement and interaction across the entire building network.